Probabilistic Modeling and Parameter Estimation

Statistical Foundations of AI and ML, Module 2

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Probabilistic Modeling of Data

- For Machine Learning, we start with data: $\{x_1, x_2, \ldots, x_N\}$
  we begin with observations
- The data may be binary, discrete, real-valued, vectors, or mixed This matters because different data types suggest different probability models.
- Our aim may be to predict the next measurement (in case of sequential data), or class-labels, or clusters in the data
  we want to predict : Predict the next value (time series), Classify something, Group Data (Clustering)
- This requires a mathematical framework to accommodate the observations Data alone is not enough. We need a model to describe variability, encode assumptions, and allow inference.
- Probabilistic Model: imagine the measurements as realizations of random variables $\{X_1, \ldots, X_N\}$
  Instead of seeing the data as fixed numbers, we imagine : x_i as a Realization of X_i
  The data we saw is one outcome.
- What distributions do these RVs follow?

Choosing a distribution is equivalent to proposing a story about how data are generated.

Many other outcomes are possible, there is an underlying random mechanism generating data

Why probabilistic model?

- Probabilistic Models allow an elegant modeling of (in)dependence between the observations
- They provide a *story* of the origin of the data, allow for generation of more data
- Provides flexibility of model by allowing complex interactions including other latent variables   Latent variables = hidden variables.
- Allow the estimation of unobserved variables using probabilistic inference   P(hidden variable | observed data)
- Allow for uncertainty analysis by specifying ranges of possible values of measurements

Instead of saying: The parameter is 5.
We say: The parameter is probably between 4.7 and 5.3.

Model Design Choices

- Observations $\{x_1, \ldots, x_N\}$ are realizations of RVs $\{X_1, \ldots, X_N\}$
- Need to consider models $P(X_1, \ldots, X_N)$
  1. Are all these RVs independent?
  2. What are their distributions?
  3. What are the parameter values of those distributions?

We must describe the joint distribution over all the variables, this will answer: How likely is any possible dataset?

- Simplest case: Independent and Identically Distributed (IID)
- $X_i \sim f(\theta)$ where f: distribution, θ: parameter values

same generating mechanism each time, No interaction, same params

- Independent but NOT identically distributed: $X_i \sim f_i(\theta_i)$, i.e. each variable has its own distribution
- NOT Independent: specify the joint or conditional distributions!

Probabilistic Model Fitting

P(x|model) is known as Likelihood,A model that assigns higher probab to observed data fits better

- All models are imagined, no **true** model
- Some models can fit the data better than other models
- Each model allows us to define *joint distribution* over RVs
- Eg. $P_1(\{X_1, \ldots, X_N\}) = \prod_{i=1}^{N} f_i(\theta_i)$ (Independent but not identically distributed)
- Or $P_2(\{X_1, \ldots, X_N\}) = g(X_1, \ldots, X_N)$ (not independent)
- The model for which $P(x_1, \ldots, x_N)$ is maximum, is said to fit the data best!
- Different models may fit different observation sets!

A model that fits one dataset may fail on another, which leads to overfitting, model selection and cross validatiion.

Fitting a model means evaluating the Joint probab : P(X1,X2,....,X_N) at the observed data.

Probabilistic Model Fitting

- Consider observations of 10 tosses $x = \{HHHHHTTTTT\}$
- Model 1 (M1): All tosses from same coin with bias 0.5
- Model 2 (M2): First 5 tosses from coin with bias 0.9, remaining tosses from another coin with bias 0.1
- $P_{M1}(x) = 0.5^{10}$, $P_{M2}(x) = 0.9^{10}$ 0.9^10 >>> 0.5^10 so from pure likelihood perspective the second model wins.
- Second model fits data better, but first model is **simpler**!
- Simpler model: fewer parameters, better generalization
- Model design: trade-off between data fit and simplicity!

Model 1 has only 1 param
Model 2 has more structure, more assumptions , more params

How to choose a Model?

- Are the observations *independent*?
- Consider the physics of the problem to get answers
- If not, carry out statistical tests of independence hypothesis (to be discussed later)
- Choose nature of distribution for X_i
- *Support* of the distribution must match that of observations
- *PDF/PMF* of the distribution must match histogram of observations
- Statistical testing of Hypothesis should be used (to be discussed later)

Parameter Estimation

- Question: What are the parameter values of the chosen distribution?
- Answer: values that cause the distribution to fit the data!
- Simplest approach: method of moments estimation (MME)
- The theoretical moments of the distribution must fit the empirical moments
- Theoretical moments: $M_t(X) = E(X^t) = \int_{Support(X)} x^t f_X(x)dx$ where f_X is the PDF of the chosen distribution (analogously for PMF)
- Empirical moments: $m_t(\{x_1, \ldots, x_N\}) = \frac{1}{N}\sum_{i=1}^{N} x_i^t$

draw samples from the experiment, the data which u have,raise to power 't'

- Form T equations like $M_t(X) = m_t(\{x_1, \ldots, x_N\})$, and solve simultaneously for the parameters of f_X

if N is large enough then empirical moment tends to theoretical moment

Common Discrete Distributions

For bernoulli distribution, expected value of RV = p , just one param so we need just one eqn to solve.

Distribution	Support	PMF	Parameters
Bernoulli	$\{0,1\}$	$p^x(1-p)^{(1-x)}$	p
Binomial	$\mathcal{Z}$	$\binom{N}{x}p^x(1-p)^{N-x}$	N,p
Poisson	$\mathcal{Z}$	$\frac{e^{-\lambda}\lambda^x}{x!}$	λ
Geometric	$\mathcal{Z}^+$	$(1-p)^{x-1}p$	p
Categorical	$\{V_1,\ldots,V_K\}$	$\prod_{k=1}^K p_k^{I(x=k)}$	$(p_1,p_2,\ldots,p_k)$
Multinomial	$\mathcal{Z}^K$	$\frac{N!}{n_1!..n_K!}\prod_{k=1}^K p_k^{n_k}$	$(N_1,p_1,\ldots,N_K,p_K)$

Common Continuous Distributions

Distribution	Support	PDF	Parameters
Beta	$(0, 1)$	$\frac{1}{B(a,b)} x^{(a-1)}(1-x)^{(b-1)}$	(a, b)
Gamma	$\mathcal{R}+$	$\frac{\beta^{\alpha}}{\Gamma(\alpha)} x^{\alpha-1} exp(-\beta x)$	(α, β)
Gaussian	$\mathcal{R}$	$\frac{1}{\sqrt{(2\pi)}\sigma} exp(-\frac{1}{2\sigma^2}(x-\mu)^2)$	(μ, σ)
M.V. Gaussian	$\mathcal{R}^D$	$\frac{1}{2\pi^{\frac{D}{2}} \lvert\Sigma\rvert^{\frac{1}{2}}} exp(-\frac{1}{2}(x-\mu)^T \Sigma^{-1}(x-\mu)$	(μ, Σ)

Maximum Likelihood Estimation (MLE)

- Choose the parameter values such that joint probability/PDF of the observations is maximized!
- i.e. Choose parameters so that model fits the data best!
- Write down joint PMF/PDF of the observations
- $prob(X_1 = x_1, X_2 = x_2, \dots, X_N = x_N) = \prod_{n=1}^{N} prob(X_n = x_n)$
- This is also called **likelihood function** $\mathcal{L}(p)$ of parameters p of the distribution (prob)
- Choose parameters such that this likelihood is maximized!
- Differentiate w.r.t p, equate to 0, solve equations!

MLE of Bernoulli

- Input: results of N tosses, $X_i \in \{0, 1\}$
- Based on support, we choose Bernoulli Distribution
- Need to estimate parameter p

$$\mathcal{L}(p) = \prod_{n=1}^{N} prob(X_n = x_n) = \prod_{n=1}^{N} p^{x_n}(1-p)^{1-x_n} = p^{N_1}(1-p)^{N_0} \quad (1)$$

$$p_{MLE} = argmax_p \mathcal{L}(p) = \frac{N_1}{N_1 + N_0} \quad (2)$$

MLE for Poisson

- Input: N integer observations, $X_i \in \mathcal{Z}$
- Based on support and histogram, we may choose Poisson Distribution
- Need to estimate parameter λ

$$\mathcal{L}(\lambda) = \prod_{n=1}^{N} prob(X_n = x_n) \propto \prod_{n=1}^{N} e^{-\lambda}\lambda^{x_n} = e^{-N\lambda}\lambda^{\sum_{n=1}^{N} x_n} \tag{3}$$

$$\lambda_{MLE} = argmax_{\lambda}\mathcal{L}(\lambda) = \frac{\sum_{n=1}^{N} x_n}{N} \tag{4}$$

MLE for Gaussian

- Input: N real observations, $X_i \in \mathcal{R}$
- Based on support and histogram, we may choose Gaussian/Normal Distribution
- Need to estimate parameter (μ, σ)

$$\begin{aligned} \mathcal{L}(\mu, \sigma) = \prod_{n=1}^{N} prob(X_n = x_n) & \propto \prod_{n=1}^{N} \frac{1}{\sigma} exp(-\frac{(x_n - \mu)^2}{\sigma^2}) \\ & = \frac{1}{\sigma^N} exp(-\sum_{n=1}^{N} \frac{(x_n - \mu)^2}{\sigma^2}) \end{aligned} \quad (5)$$

$$\mu_{MLE}, \sigma_{MLE} = argmax_{\mu,\sigma} \mathcal{L}(\mu, \sigma)$$

$$\mu_{MLE} = \frac{\sum_{n=1}^{N} x_n}{N}, \sigma_{MLE} = \frac{1}{N} \sum_{n=1}^{N} (x_n - \mu_{MLE})^2 \quad (6)$$

MLE for Multivariate Gaussian

- Input: N real vector observations, $X_i \in \mathcal{R}^D$
- Based on support and histogram, we may choose Gaussian/Normal Distribution
- Need to estimate parameter (μ, Σ)

$$\mathcal{L}(\mu, \Sigma) = \prod_{n=1}^{N} prob(X_n = x_n) \quad \propto \quad \prod_{n=1}^{N} \frac{1}{|\Sigma|^{\frac{1}{2}}} exp(-(x_n - \mu)^T \Sigma^{-1} (x_n - \mu))$$

$$= \frac{1}{|\Sigma|^{\frac{N}{2}}} exp(-\sum_{n=1}^{N} (x_n - \mu)^T \Sigma^{-1} (x_n - \mu)) \quad (7)$$

$$\mu_{MLE}, \Sigma_{MLE} = argmax_{\mu,\Sigma} \mathcal{L}(\mu, \Sigma)$$

outer product :

$$\mu_{MLE} = \frac{\sum_{n=1}^{N} x_n}{N}, \Sigma_{MLE} = \frac{1}{N} \sum_{n=1}^{N} (x_n - \mu_{MLE})(x_n - \mu_{MLE})^T \quad (8)$$

Drawbacks of Maximum Likelihood

- Gives a single optimal value of parameters instead of feasible range
- Can give unreliable results when data is less
- Even with a lot of data, can it give *correct* estimate?
- Toy experiment: Generate 1000 samples from a Gaussian with known parameters μ, σ^2
- Select a subset S1 of 100 samples, and estimate μ as the sample mean
- Repeat this with another sample S2. Do the estimates of μ match?
- If they don't match, how can we expect to find the right μ by Maximum-Likelihood?

Properties of Estimator

- Consider T_N, a function of N Random Variables $T_N = f(X_1, \ldots, X_N)$
- T_N is called a *statistic* of $\{X_i\}$
- If it is used to *estimate* any parametric function $g(\theta)$, we call T_N an *estimator* of $g(\theta)$
- For example, sample mean $\bar{X} = \frac{1}{N}\sum_{i=1}^{N} X_i$ is an *estimator* of μ, assuming $X_i \sim \mathcal{N}(\mu, \sigma^2)$
- T_N is a random variable with associated probability distribution
- *Bias of estimation*: $E(T_N - g(\theta))$
- If $E(T_N - g(\theta)) = 0 \forall \theta$, we say T_N is an *unbiased estimator* of $g(\theta)$ what is meaning of for all theta here?

Properties of Estimator

- Every single value of T_N need not equal $g(\theta)$, even if T_N is unbiased estimator
- In the Gaussian sample experiment, we need to take the expected value of the different sample means!
- If $Lim_{N\to\infty} E(T_N - g(\theta)) = 0$, we say T_N is *asymptotically unbiased*
- i.e. Given infinite data, we can get unbiased estimate
- *Consistent Estimator*: $Lt_{N\to\infty} P(|T_N - g(\theta)| < \epsilon) = 1$, $\forall\theta$ and $\forall\epsilon > 0$
- i.e. Every individual estimate is arbitrarily close to $g(\theta)$, if based on large enough sample
- *Mean Square Error*: $E((T_N - g(\theta))^2)$

Properties of Maximum Likelihood

- Maximum Likelihood Estimate is always a Consistent Estimator
- Maximum Likelihood Estimate need not be an unbiased Estimator
- Maximum Likelihood Estimator for Gaussian Mean parameter is unbiased
- Maximum Likelihood Estimator for Gaussian Variance parameter is asymptotically unbiased
- Unbiased estimator of Gaussian Variance $\frac{1}{N-1}\sum_{i=1}^{N}(X_i - \bar{X})^2$
- However, the MLE has less Mean Square Error than the unbiased estimate!

Bayesian Parameter Estimation

- Maximum likelihood estimate - entirely based on data
- But if data is not reliable?
- Bayesian approach: we may have some prior beliefs
- Bayesian approach: combine our prior beliefs with evidence, i.e. data
- Bayesian approach: keep updating our beliefs as more and more data comes in!

Bayesian Parameter Estimation

- Consider the parameters as random variables
- Put a prior distribution on the parameters
- $posterior(param|data) \propto prob(data|param) * prior(param)$
- $prob(data|param) = \mathcal{L}(param)$ (likelihood function)
- Difference from MLE - we get a distribution of the parameter instead of single value
- Maximum A-Posteriori (MAP) estimate: $param_{MAP} = argmax_{param} posterior(param|data)$
- Expected value of parameter from posterior: $param_{Bayes} = E(param|data)$

Bayesian Parameter Estimation

- How to choose prior distribution?
  - Reflect our belief on parameter
  - Mathematical tractability (posterior should be a valid distribution)
- Some likelihood functions have conjugate prior
- Prior and Posterior on parameters should be same distribution with different parameters!
  - Easy to interpret

Bayesian estimate of Bernoulli Distribution

- Data: $\{x_1, \ldots, x_N\} \in \{0,1\}$, Model: $X_i \sim Bernoulli(p)$
- $p \in (0,1)$, so $prior(p) = Beta(a,b)$
  - (a,b) hyperparameters - parameters of prior
  - Assume $a = b$ if no information

$$\begin{aligned} posterior(p|X) &\propto \prod_{i=1}^{N} prob(X_i = x_i|p) * prior(p) \\ &= p^{N_1}(1-p)^{N_0} * p^{a-1}(1-p)^{b-1} \\ &= p^{N_1+a-1}(1-p)^{N_0+b-1} \end{aligned} \tag{9}$$

- $posterior(p) : Beta(N_1 + a, N_0 + b)$
- $p_{Bayes} = \frac{N_1+a}{N+a+b}$ (expection), $p_{MAP} = \frac{N_1+a-1}{N+a+b-2}$ (mode)

Bayesian estimate of Bernoulli parameters

- $prior(p) = Beta(5, 7)$, $p_{MAP} = 4/10$
- $X_1 = TAIL$, $posterior(p) = Beta(5, 8)$, $p_{MAP} = 4/11$
- $X_2 = HEAD$, $posterior(p) = Beta(6, 8)$, $p_{MAP} = 5/12$
- $X_3 = HEAD$, $posterior(p) = Beta(7, 8)$, $p_{MAP} = 6/13$
- $X_4 = TAIL$, $posterior(p) = Beta(7, 9)$, $p_{MAP} = 6/14$
- $X_5 = HEAD$, $posterior(p) = Beta(8, 9)$, $p_{MAP} = 7/15$
- $X_6 = HEAD$, $posterior(p) = Beta(9, 9)$, $p_{MAP} = 8/16$

Bayesian estimate of Gaussian Distribution - variance known

- Data: $\{x_1, \ldots, x_N\} \in \mathcal{R}$, Model: $X_i \sim \mathcal{N}(\mu, \sigma^2)$
- $\mu \in \mathcal{R}$, so $prior(\mu) = \mathcal{N}(\mu_0, \sigma_0^2)$
- Assume σ is known for simplicity

$$\begin{aligned} posterior(\mu|X) &\propto \prod_{i=1}^{N} prob(X_i = x_i|\mu) * prior(\mu) \\ &= \frac{1}{\sigma^N} exp(-\frac{\sum_{i=1}^{N}(x_i - \mu)^2}{2\sigma^2}) * \frac{1}{\sigma_0} exp(-\frac{(\mu - \mu_0)^2}{2\sigma_0^2}) \\ &= \mathcal{N}(\frac{\frac{N}{\sigma^2}\hat{X} + \frac{\mu_0}{\sigma_0^2}}{\frac{N}{\sigma^2} + \frac{1}{\sigma_0^2}}, \frac{1}{\frac{N}{\sigma^2} + \frac{1}{\sigma_0^2}}) \qquad (10) \end{aligned}$$

where $\hat{X} = \frac{1}{N}\sum_{i=1}^{N} X_i$

Bayesian estimate of Gaussian Distribution - Variance Unknown

- Data: $\{x_1, \ldots, x_N\} \in \mathcal{R}$, Model: $X_i \sim \mathcal{N}(\mu, \tau)$ where $\tau = \frac{1}{\sigma^2}$
- Define $prior(\mu, \tau) = prior_1(\mu|\tau), prior_2(\tau)$
- $prior_1(\mu|\tau) = \mathcal{N}(\mu_0, \eta\tau)$, $prior_2(\tau) = Gamma(a, b)$
- $posterior(\mu, \tau) = \mathcal{L}(\mu, \tau, X) * prior_1(\mu|\tau) * prior_2(\tau)$
- $posterior(\mu) = \int posterior(\mu, \tau)d\tau : GaussianDistribution$
- $posterior(\tau) = \int posterior(\mu, \tau)d\mu : GammaDistribution$

Confidence Interval of Estimation

- Sometimes we may not be able to have a point estimate of parameter θ
- Consider two statistics $U(X)$ and $L(X)$, within which we expect the target θ to lie
- If $P_\theta(\theta \in (L(X), U(X))) = 1 - \alpha$, then $(L(X), U(X))$ is the $100(1-\alpha)\%$ confidence interval for θ
- Example: $X_i \sim \mathcal{N}(\mu, \sigma^2)$, then $\bar{X} \sim \mathcal{N}(\mu, \frac{\sigma^2}{n})$
- So, $P(\mu - \frac{\sigma}{\sqrt{n}} Z_{\alpha/2} < \bar{X} < \mu + \frac{\sigma}{\sqrt{n}} Z_{\alpha/2}) = 1 - \alpha$, where $\Phi(-Z_{\alpha/2}) = \alpha/2$ and $\Phi(Z_{\alpha/2}) = 1 - \alpha/2$
- Thus $(\bar{X} - \frac{\sigma}{\sqrt{n}} Z_{\alpha/2}, \bar{X} + \frac{\sigma}{\sqrt{n}} Z_{\alpha/2})$ is the $100(1-\alpha)\%$ confidence interval for μ

Naive Bayes Classifier as Parameter Estimation

- Dataset: $\{x_i, y_i\}_{i=1}^N$ where $y_i \in \{1, \ldots, C\}$
- x_i has D discrete-valued features x_{ij}, s.t. $x_{ij} \in \{1, \ldots, M_j\}$
- Bayesian Classifier: X, Y jointly distributed random variables!
- Prior distribution on label: $Y \sim Cat(\pi)$ $(\pi = \{\pi_1, \ldots, \pi_C\})$
- Naive-Bayes model: $P(X|Y) = \prod_{j=1}^D P(X_j|Y)$
- Class-conditional distribution: $P(X_j|Y = c) \sim Cat(\theta_{cj})$ (parameters specific to feature and class label)
- Joint distribution $P(X, Y) = \prod_{i=1}^N P(X_i, Y_i) = \prod_{i=1}^N P(X_i|Y_i)P(Y_i)$
- Substituting the PMFs, $P(X, Y) =$ $\prod_{i=1}^N \prod_{c=1}^C \prod_{j=1}^D \prod_{k=1}^{M_j} \theta_{cjk}^{I(X_{ij}=k)I(Y_i=c)} \prod_{i=1}^N \prod_{c=1}^C \pi_c^{I(Y_i=c)}$

Naive Bayes Classifier with Dirichlet Prior

- Maximum-likelihood estimate of θ_{cj} and π: MLE of categorical
- $\pi_{MLE}(c) = \frac{\sum_{i=1}^{N} I(Y_i=c)}{\sum_{i=1}^{N}\sum_{c=1}^{C} I(Y_i=c)} = \frac{N_c}{N}$ (fraction of examples from class c)
- $\theta_{MLE}(cjk) = \frac{\sum_{i=1}^{N} I(X_{ij}=k)I(Y_i=c)}{\sum_{i=1}^{N}\sum_{k=1}^{M_j} I(X_{ij}=k)I(Y_i=c)} = \frac{N_{cjk}}{N_c}$ (fraction of examples in class c for which j-th feature has value k)
- For Bayesian analysis, we may put Dirichlet priors
- $\pi \sim Dir(\alpha)$ and $\theta_{cj} \sim Dir(\beta_{cj})$
- $P(\theta, \pi|\mathcal{D}) \propto P(\mathcal{D}|\theta, \pi)P(\theta)P(\pi)$ where $\mathcal{D} = \{x_i, y_i\}_{i=1}^{N}$
- $P(\pi|\mathcal{D}) = Dir(N_1 + \alpha_1, \ldots, N_C + \alpha_C)$
- $P(\theta_{cj}|\mathcal{D}) = Dir(N_{cj1} + \beta_{cj1}, \ldots)$

Linear Regression as Parameter Estimation

- Data $\{x_i, y_i\}_{i=1}^N$ where $X_i \in \mathcal{R}^D$ and $Y_i \in \mathcal{R}$
- Linear Regression model: $y_i = W^T x_i$ where $W \in \mathcal{R}^D$
- Ordinary Least Square estimate: $W_{OLS} = argmin_W \sum_{i=1}^N (y_i - W^T x_i)^2$
- Probabilistic model: $Y_i = W^T x_i + \epsilon_i$, where $\epsilon_i \sim \mathcal{N}(0, \sigma^2)$ random noise
- Equivalently, $Y_i \sim \mathcal{N}(W^T x_i, \sigma^2)$ Y_i is the RV
- The OLS estimate is equal to Maximum Likelihood estimate!
- But can we use prior knowledge about W to encourage structural properties?

Ridge Regression

- Knowledge: W is centered around origin, most likely within a sphere of radius 3ρ
- Prior $W \sim \mathcal{N}(0^D, \rho^2 I)$ where I is $D \times D$ identity matrix and 0^D is D-dim zero-vector
- In general, $W \sim \mathcal{N}(W_0, V_0)$
- Posterior distribution on W: $p(W|\mathcal{D}) = p(\mathcal{D}|W)p(W)$
- Note: Likelihood $p(\mathcal{D}|W)$ and prior $p(W)$ are both Gaussian!
- Multiplying PDFs, posterior PDF $\mathcal{N}(W_N, V_N)$ where $W_N = V_N V_0^{-1} W_0 + \frac{1}{\sigma^2} V_N \sum_{i=1}^N X_i^T Y_i$, $V_N = (V_0^{-1} + \frac{1}{\sigma^2} \sum_{i=1}^N X_i X_i^T)^{-1}$

WE can put some guassian prior over W and posterior will again turn out to be guassian

Prediction using Linear Regression

- Aim: given training data $\mathcal{D}$, make predictions on test data x
- $p(Y|X,\mathcal{D}) = \int p(Y|X,W)p(W|\mathcal{D})dW$ (last term is the posterior)
- $p(Y|X,W) = \frac{1}{\sqrt{(2\pi)}\sigma} exp(-\frac{1}{2\sigma^2}(Y - W^T X)^2)$, $p(W|\mathcal{D}) = \frac{1}{(2\pi)^{D/2}|V_N|^{1/2}} exp(-0.5(W - W_N)^T V_N^{-1}(W - W_N))$
- **Theorem**: If $y|x \sim \mathcal{N}(Ax + b, \Sigma)$ and $x \sim \mathcal{N}(\mu_x, \Sigma_x)$, then $x|y \sim \mathcal{N}(\mu_{x|y}, \Sigma_{x|y})$ and $y \sim \mathcal{N}(\mu_y, \Sigma_y)$
- Here, $\mu_{x|y} = \Sigma_{x|y}(A^T \Sigma^{-1}(y - b) + \Sigma_x^{-1}\mu_x)$, $\Sigma_{x|y} = (\Sigma_x^{-1} + A^T \Sigma^{-1} A)^{-1}$, $\mu_y = A\mu_x + b$, $\Sigma_y = \Sigma + A\Sigma_x A^T$
- Thus, $Y|X,\mathcal{D} \sim \mathcal{N}(W_N^T X, \sigma^2 + X^T V_N X)$

Sparse Regression with Slab-and-Spike Prior

- Suppose we want W to be a sparse vector
- $W = V \circledast \eta$, where η is a binary vector that indicates which elements of W are 0 (slabs and spikes)
- Then $V \sim \mathcal{N}(0^D, \rho^2 I)$, and $\eta_d \sim Ber(\eta_0)$
- Posterior $p(W|\mathcal{D}) \propto p(\mathcal{D}|W)p(V)p(\eta)$
- The log-PDF: $-\frac{1}{2\sigma^2}\sum_{i=1}^{N}(y_i - \sum_{j=1}^{D}\eta_j V_j X_{ij})^2 - \frac{1}{2\rho^2}\sum_j V_j^2 + (\sum_{j=1}^{D}\eta_j) log(\eta_0) + (D - (\sum_{j=1}^{D}\eta_j)) log(1-\eta_0)$ where $\sum_{j=1}^{D}\eta_j$ is the number of non-zero elements
- Clearly, the posterior does not belong to any standard distribution family
- MAP estimate: maximize the posterior by co-ordinate ascent approach (alternating maximization) over η_j and V_j

Sparse Regression with LASSO

- $W_{sparse} = argmin_W \sum_{i=1}^{N} (y_i - W^T X_i)^2 + \lambda ||W||_0$
- Non-convex, so we relax
  $W_{lasso} = argmin_W \sum_{i=1}^{N} (y_i - W^T X_i)^2 + \lambda ||W||_1$
- LASSO (least absolute shrinkage and selection operator) works as below

Algorithm 13.1: Coordinate descent for lasso (aka shooting algorithm)

Initialize $\mathbf{w} = (\mathbf{X}^T\mathbf{X} + \lambda\mathbf{I})^{-1}\mathbf{X}^T\mathbf{y}$;
repeat
  for $j = 1, \ldots, D$ **do**
    $a_j = 2\sum_{i=1}^{n} x_{ij}^2$;
    $c_j = 2\sum_{i=1}^{n} x_{ij}(y_i - \mathbf{w}^T\mathbf{x}_i + w_j x_{ij})$;
    $w_j = \text{soft}(\frac{c_j}{a_j}, \frac{\lambda}{a_j})$;

Sparse Regression with Laplace Prior

- Bayesian analogy: consider a Laplace prior on W
- $Lap(W_j|0, \frac{1}{\gamma}) = \frac{\gamma}{2} exp(-\gamma|W_j|)$ (encourages W_j to be 0)
- Posterior cannot be computed directly, so need to use alternative expression of PDF called Gaussian Scale Model
- Joint PDF $p(\mathcal{D}, W) \propto exp(-\frac{1}{2}\sum_{i=1}^{N}(y_i - W^T x_i)^2) exp(-\frac{1}{2} W^T D_\lambda W)$
- Here, D_λ is a diagonal matrix
- In general, $X \sim Lap(\mu, \beta) \rightarrow f_X(x) = \frac{1}{2\beta} exp(-\frac{|x-\mu|}{\beta})$
- μ analogous to Gaussian Mean/Mode, β analogous to Gaussian variance

Autoregressive Models

- What if the observations cannot be considered as independent?
- Factorize as $p(X) = p(X_1)p(X_2|X_1)\dots p(X_i|X_1,\dots,X_{i-1})\dots$
- Define $X_i = f(X_1,\dots,X_{i-1})$ where f is a suitable stochastic function
- Autoregressive model of order k: $X_t \sim g(W_1X_{t-1} + \dots + W_kX_{t-k}) + \epsilon_t$ where $\epsilon_t \sim \mathcal{N}(0,\sigma^2)$
- If g is identity function, then $X_t \sim \mathcal{N}(W^TX_t^k, \sigma^2)$ where $X_t^k = \{X_{t-1},\dots,X_{t-k}\}$
- Parameters may be estimated by Maximum-Likelihood (how?)